ACADEMIC YEAR 2026

A RISK-CALIBRATED ADAPTIVE HANDOVER METHOD FOR TEACHER-GUIDED FLOW-MATCHING TEXT-TO-SPEECH INFERENCE



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PRESENTATION STRUCTURE

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★ Contribution and summary

App. Preliminary tests and supplementary data

SECTION 1.1 · BACKGROUND OF THE STUDY

FLOW-MATCHING TTS: COST & DISTILLATION TRADE-OFFS

FLOW MATCHING

$$dx/dt = v_\theta(x,\ t)$$

Noise moves to a mel-spectrogram step by step; **each step is one network evaluation**. F5-TTS is our teacher.

CLASSIFIER-FREE GUIDANCE (CFG)

$$v = v_u + w(v_c - v_u)$$

Conditional + unconditional prediction: **2 forward passes** per guided teacher step.

FEW-STEP STUDENT DISTILLATION

4 steps

DMOSpeech distilled the teacher into a **few-step student**, 128 → 4.

MODE SHRINKAGE & HYBRID RECOVERY

The student's prosody is less varied (**mode shrinkage**). DMOSpeech 2 lets the teacher run the first steps and the student finish.

Released sampler, switching time 0.25 — teacher 18 — student 3

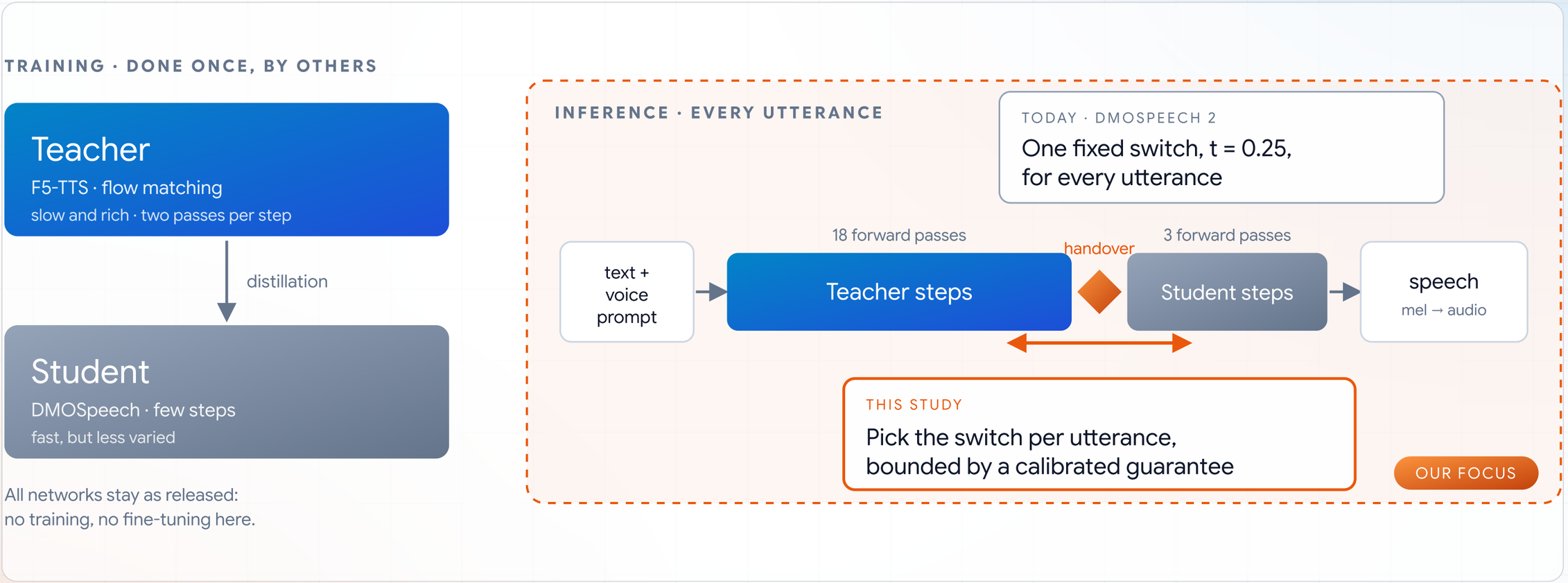


21 forward passes for **every** utterance

SOURCES Lipman et al. (2023) · Y. Chen et al. (2025), *F5-TTS* · Ho & Salimans (2022), *Classifier-free diffusion guidance* · Li et al. (2025), *DMOSpeech*, ICML · Li et al. (2026), *DMOSpeech 2*, AAAI

SECTION 1.1 · THE MODEL AT A GLANCE

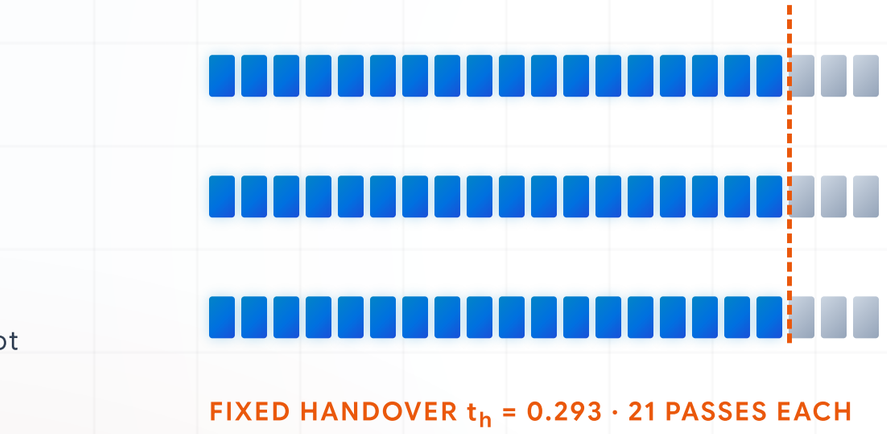
TEACHER-GUIDED TTS, AND WHERE WE WORK



SOURCES Y. Chen et al. (2025), *F5-TTS* · Li et al. (2025), *DMOSpeech* · Li et al. (2026), *DMOSpeech 2* · Proposal Sections 1.1 and 3.2 (simplified outline)

SECTION 1.2 · PROBLEM STATEMENT

LIMITATIONS OF STATIC HANDOVER SCHEDULES ACROSS INPUTS



Too late wastes guided passes. Too early may degrade speech. **Does every utterance need 18 teacher passes?**

PRELIMINARY T4 EVIDENCE · APPENDIX D.1 · 1,000 INSTANCES

Fixed early switches fail on **25–33%** of inputs (noisy prompts 49–64%, clean 1.0–2.4%). A per-input choice in hindsight needs **14.85 of 21** passes (−29%). No single early switch is safe for every input.

1.2.1 **Uncharacterized effects.** How the switching time changes WER, SIM, UTMOS and cost per utterance is unmeasured.

1.2.2 **The handover boundary.** Can inference-time signals pick an earlier switch without losing quality?

1.2.3 **No reliability guarantee.** Adaptive selection alone says nothing about the rate of added failures.

1.2.4 **Application constraints.** Faster synthesis is not evidence of benefit to autistic users.

SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Ho & Salimans (2022) · Angelopoulos et al. (2025), *Learn then test*, Ann. Appl. Stat. · Proposal Appendix D.1 (development evidence)

SECTION 1.3 · OBJECTIVES OF THE STUDY

GENERAL & SPECIFIC RESEARCH OBJECTIVES

1

CHARACTERIZE

Measure how WER, SIM, UTMOS, forward passes and runtime vary across candidate switching times and inputs.

→ QUALITY–COST MAP

2

FORMULATE

Pose selection as constrained cost minimization; design a training-free policy with no extra network evaluations; prove its properties.

→ POLICY + 4 PROPOSITIONS

3

CALIBRATE

Use Learn-Then-Test to keep added-failure risk within a tolerance α at confidence $1 - \delta$; check it on held-out speakers.

→ CERTIFIED THRESHOLD $\hat{\lambda}$

4

COMPARE

Benchmark SNR-only, stabilization-only and combined variants against fixed switches and an oracle, clean and noisy.

→ DOES EACH SIGNAL ADD VALUE?

FROM Proposal, Section 1.3 · all pretrained weights frozen; training-free

SECTIONS 1.4–1.5 · SCOPE & SIGNIFICANCE

SCOPE, DELIMITATIONS, & TARGET BENEFICIARIES

IN **Only the switching time varies.** Restart level fixed at the released default (j = 1) as a controlled setting.

IN Teacher, student, duration predictor and vocoder **stay frozen**.

IN English zero-shot prompts from **Seed-TTS test-en**, clean and 15 dB noisy.

IN All runs on an **NVIDIA Tesla T4** through Google Colab.

OUT A per-utterance guarantee; the bound holds **on average** over inputs.

OUT Building or testing an assistive app; clinical or learner outcomes.

WHO BENEFITS

SPEECH RESEARCHERS

A formulation, proofs and calibration for teacher-guided samplers in general.

TTS DEVELOPERS

Lower cost with a stated bound, no retraining.

ASSISTIVE-TECH DEVELOPERS

Cheaper, predictable synthesis for personalized voices.

FUTURE RESEARCHERS

A reusable protocol: disjoint calibration, fixed failure rule, exact cost.

SOURCES Anastassiou et al. (2024), *Seed-TTS* · Ardila et al. (2020), *Common Voice*, LREC · Muharib et al. (2024), JADD · Therrien et al. (2025), AAC

SECTION 2.4 · RELATED LITERATURE

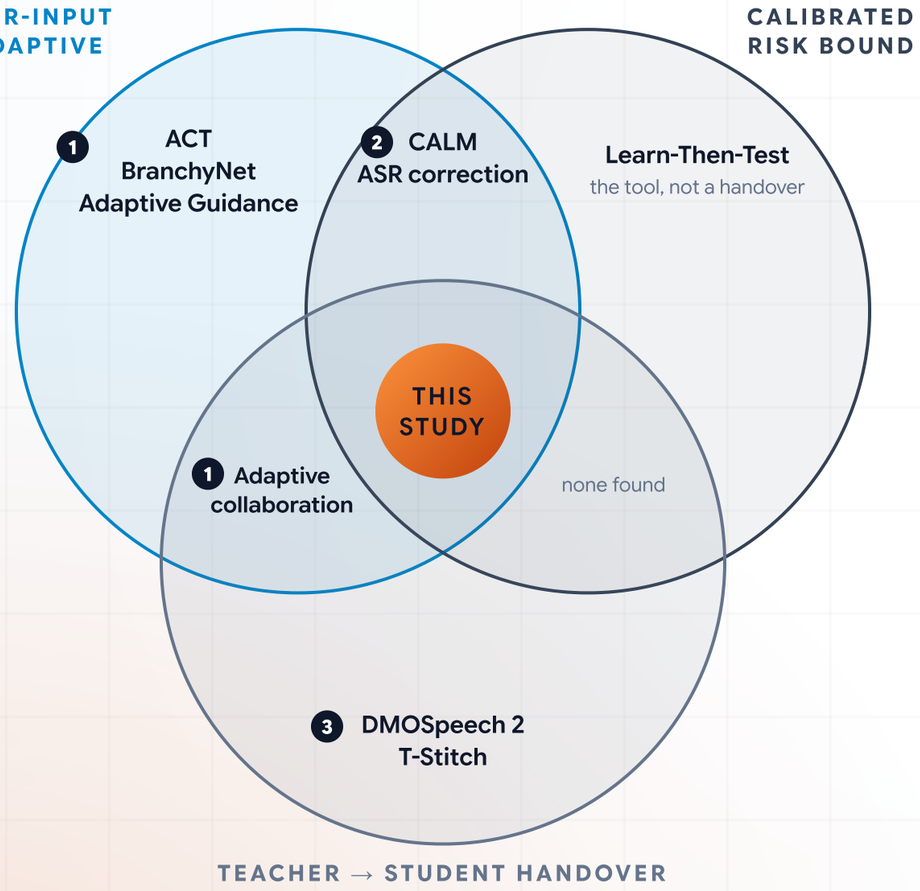
INPUT-ADAPTIVE COMPUTATION & DYNAMIC INFERENCE

METHOD	WHAT IS ADAPTED	DECISION SIGNAL	RULE	GUARANTEE
ACT Graves, 2016	Internal steps per input	Learned halting unit	Learned	NONE
BranchyNet Teerapittayanon et al., 2016	Exit layer	Classifier confidence	Tuned threshold	NONE
CALM Schuster et al., 2022	Exit layer per token	Confidence or saturation	Calibrated threshold	RISK CONTROL
Adaptive collaboration Starodubcev et al., 2024	Whether the teacher refines	Quality estimator on a full student sample	Tuned threshold	NONE
T-Stitch Pan et al., 2025	Which model runs each step	None	Preset switching point	NONE
Adaptive Guidance Castillo et al., 2025	Whether to run the unguided pass	Agreement of guided and conditional updates	Tuned threshold	NONE
Adaptive ASR correction Damri & Laufer-Goldshtein, 2026	Hypotheses per input	ASR confidence scores	Calibrated threshold	RISK CONTROL (LTT)
This study	Teacher-to-student switching time	Prompt SNR + endpoint stabilization	Calibrated threshold	RISK CONTROL (LTT)

SOURCES Graves (2016) · Teerapittayanon et al. (2016), ICPR · Schuster et al. (2022), NeurIPS · Starodubcev et al. (2024), CVPR · Pan et al. (2025) · Castillo et al. (2025) · Damri & Laufer-Goldshtein (2026) · Proposal Table 2.1

SECTIONS 2.4–2.8 · LITERATURE SYNTHESIS

WHERE PRIOR WORK LEAVES THE GAPS



1 Adaptive, but no bound on quality loss

Per-input rules built on a learned halting unit or a hand-tuned threshold. None states how much quality the saving may cost.

2 Bounded, but not for a handover

CALM and adaptive ASR correction calibrate a threshold with risk control, for exit layers and recognition hypotheses, not for a teacher-to-student switch in speech synthesis.

3 A handover, but one fixed for all inputs

DMOSpeech 2 switches at one time for every utterance; T-Stitch uses a preset switching point. Neither adapts to the input or bounds the risk.

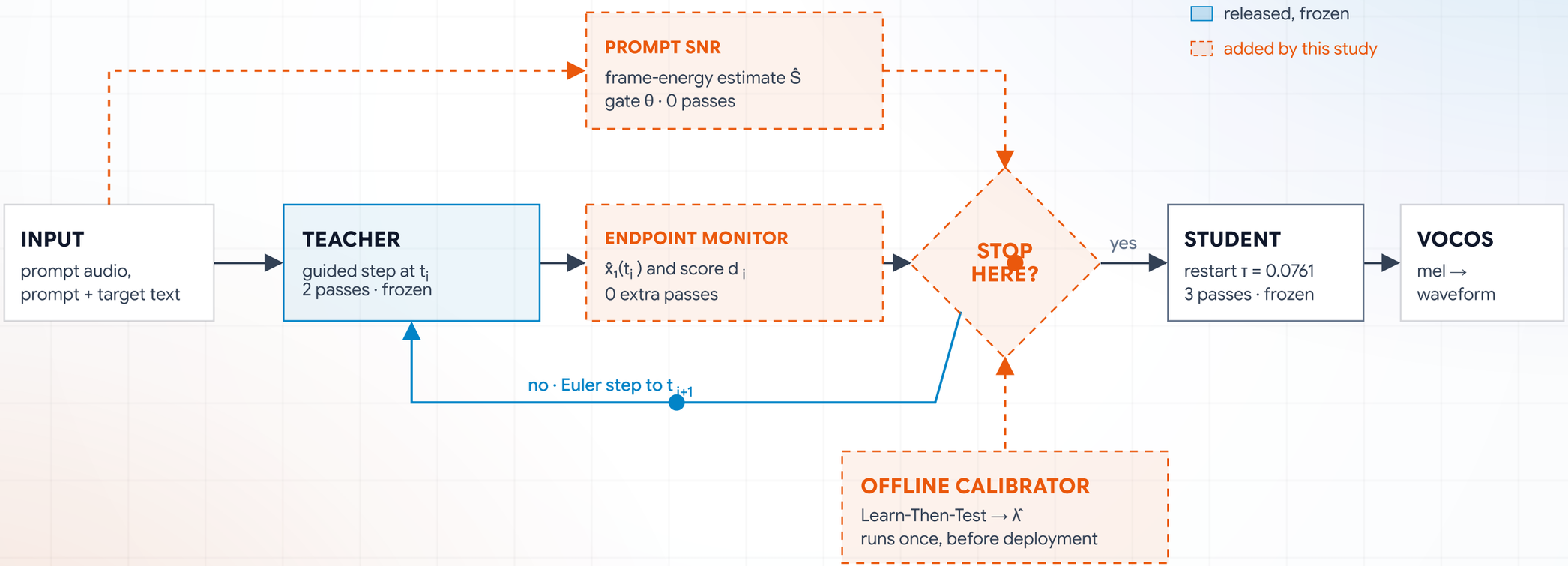
★ The centre is unoccupied

No reviewed method pairs a per-utterance handover with a calibrated bound. This describes the sources examined, not every adaptive method that exists.

SOURCES Graves (2016) · Teerapittayanon et al. (2016) · Schuster et al. (2022) · Starodubcev et al. (2024) · Pan et al. (2025) · Castillo et al. (2025) · Damri et al. (2026) · Li et al. (2026) · Angelopoulos et al. (2025) · Proposal Chapter II

SECTION 3.2 · SYSTEM ARCHITECTURE

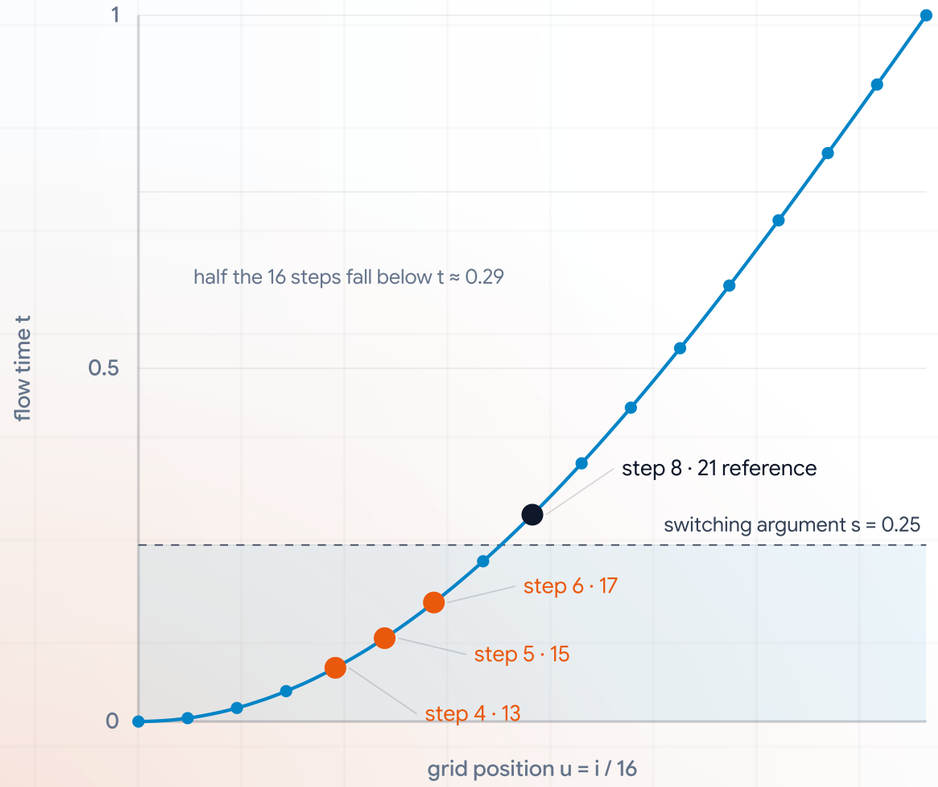
MODULAR ADAPTIVE HANDOVER ARCHITECTURE



SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Y. Chen et al. (2025), *F5-TTS* · Siuzdak (2024), *Vocos*, ICLR · Proposal Figure 3.2

SECTION 3.5 · CANDIDATE SWITCHING TIMES

SWAY SAMPLING DISCRETIZATION & CANDIDATE HANDOVER BOUNDARIES



Sway sampling packs teacher steps near $t = 0$. The released code stops at the **first grid point above** s. Cost: $C(s) = 2(k + 1) + 3$.

s = 0.07
step 4 · t = 0.076



13
−8 · 38%

s = 0.10
step 5 · t = 0.118



15
−6 · 29%

s = 0.15
step 6 · t = 0.169



17
−4 · 19%

s = 0.25
step 8 · reference



21
baseline

$$x_\tau = (1 - \tau)\,\varepsilon + \tau\,\tilde{x}, \quad \tau = 0.0761$$

student always re-noises the teacher's estimate to the same level

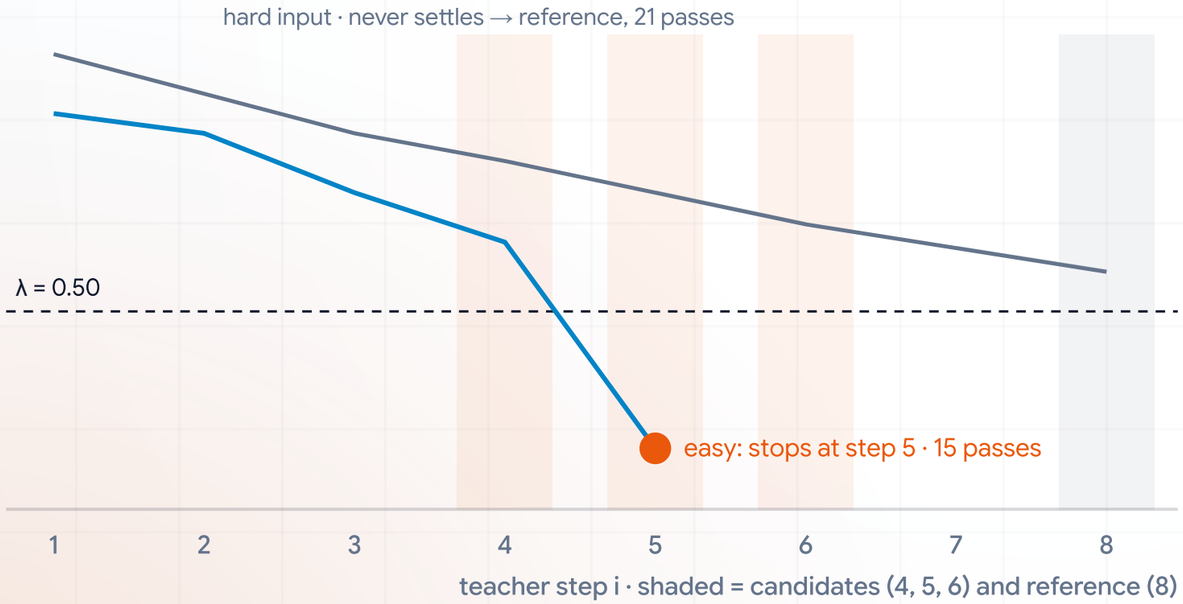
SOURCES Y. Chen et al. (2025), *F5-TTS*, ACL · Li et al. (2026), *DMOSpeech 2*, AAAI · Meng et al. (2022), *SDEdit*, ICLR · Xu et al. (2023), NeurIPS

SECTION 3.6.2 · DECISION FEATURES

ENDPOINT STABILIZATION & PROMPT ACOUSTIC QUALITY SIGNALS

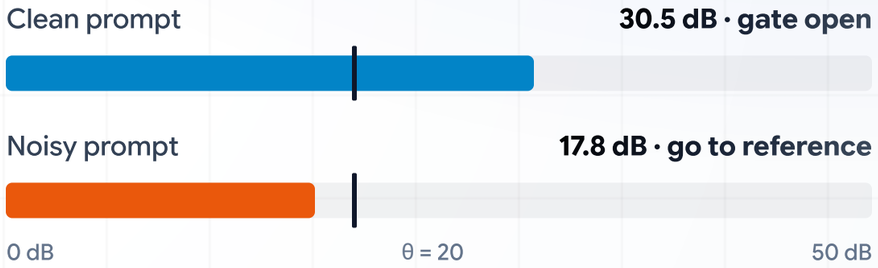
$$d_i = \| \hat{x}_1(t_i) - \hat{x}_1(t_{i-1}) \|_M \,/\, (\| \hat{x}_1(t_i) \|_M + \epsilon), \quad \hat{x}_1 = x_t + (1 - t)\, v$$

endpoint-stabilization score · small means the teacher's guess has settled



$$\hat{S} = 10 \log_{10}(\bar{E}_{\text{top}} \,/\, \bar{E}_{\text{bottom}})$$

prompt SNR, from frame energies · gate θ = 20 dB



Development diagnostics suggest the **SNR may carry most of the benefit**. The stabilization score is kept as a hypothesis to test, not an assumption.

SOURCES Lipman et al. (2024), *Flow matching guide and code* · Li et al. (2026) · Proposal Def. 3.2–3.3, Appendix A.4 and D.4 (illustrative values)

SECTION 3.6 · MATHEMATICAL FORMULATION

FORMULATION OF THE CONSTRAINED STOPPING POLICY

$$\min_{\pi} \mathrm{E}[\ C(\pi(x))\ ] \quad \text{s.t.} \quad R(\pi) = \mathrm{E}[\ f(x, \pi(x))\ ] \leq \alpha$$

cheapest policy whose added-failure risk stays under α

$$\pi_{\lambda}(x) = \text{first candidate } i_r \text{ with } d_{i_r} \leq \lambda \text{ and } \hat{S} \geq \vartheta, \text{ else reference}$$

walks the teacher steps in order · never later than step 8

ADDED FAILURE · FIXED BEFORE CALIBRATION

$f = 1$ if WER > max(**12.5%**, $\mathrm{WER}_{\mathrm{ref}}$ + **1 pt**) or SIM < $\mathrm{SIM}_{\mathrm{ref}}$ − **0.02**, on the same seeds.

Setting λ = ∞ gives the SNR-only rule; θ = −∞ gives stabilization only. **One family, one calibration.**

PROP. 3.2

ZERO OVERHEAD

Signals reuse the velocity already computed. No extra forward pass.

PROP. 3.3

MONOTONE COST

A smaller λ never stops earlier, so cost rises as λ falls.

PROP. 3.4

SAFE FALLBACK

Never more than 21 passes. No certified λ → exact reference output.

PROP. 3.5

ORACLE BOUND

No zero-risk policy beats the cheapest non-failing candidate per input.

FROM Proposal Eq. 3-11, 3-12, 3-15 and Propositions 3.2–3.5 · WER: Radford et al. (2023) · SIM: S. Chen et al. (2022)

SECTION 3.7 · RISK CALIBRATION PROTOCOL

RISK-CALIBRATED THRESHOLD SELECTION VIA LEARN-THEN-TEST

SPEAKER-DISJOINT SPLITS · SEED-TTS TEST-EN · ONE UTTERANCE PER SPEAKER

Development	Calibration	Test
100	200	200

ILLUSTRATIVE · n = 100, α = 0.05, δ = 0.10 · p = P(Bin(n, α) ≤ failures)

g = 1 · conservative

λ 0.40

failures	0
p	0.0059

≤ Δ · CERTIFY

g = 2

λ 0.50

failures	1
p	0.0371

≤ Δ · CERTIFY

g = 3

λ 0.60

failures	3
p	0.2578

> Δ · STOP

g = 4 · aggressive

λ 0.70

failures	6
p	—

NOT TESTED

λ̂ = 0.50

Guarantee: **P(R(λ̂) ≤ α) ≥ 1 − δ**, on average over new speakers. With zero failures, certifying α = 0.05 at δ = 0.10 needs **45** inputs (Hoeffding: ≈461). Protocol fixed in advance: grid 0.20–1.20 in steps of 0.05; verified on the **held-out test split** and **100 re-splits**.

SOURCES Angelopoulos et al. (2025), *Learn then test* · Bauer (1991), Stat. Med. · Clopper & Pearson (1934) · Hoeffding (1963) · Proposal Alg. 3.4, Appendix A.6

SECTIONS 3.1, 3.3–3.8 · RESEARCH DESIGN & PROTOCOL

EXPERIMENTAL WORKFLOW, BASELINES, & EVALUATION METRICS

SETUP
ALL SPLITS

1 Speaker splits
2 Clean + 15 dB noisy
3 Check seed, fallback, pass count

OBJECTIVE 1
DEVELOPMENT · EXPLORATORY

4 One teacher run, all candidates
5 Score + failures
6 Quality–cost map, fix θ

OBJECTIVE 2
NO DATA

7 Problem, policy, proofs

OBJECTIVE 3
CALIBRATION, TEST · CONFIRMATORY

8 Fix α, δ, grid
9 Binomial tests in order
10 Lock $\hat{\lambda}$ or fall back
11 Held-out + 100 re-splits

OBJECTIVE 4
TEST · CONFIRMATORY

12 Run locked variants
13 Paired comparisons
14 Noise shift, n = 50/100/200
15 Report all, incl. negative

COMPARED METHODS

Fixed s ∈ {0.07, 0.10, 0.15, **0.25**} · SNR-only · SNR + fixed later step · stabilization only · **full policy** · cost-matched random · oracle

MEASURES

WER Whisper-large-v3 · **SIM** WavLM-Base+ SV · **UTMOS** · exact **forward passes** · runtime and RTF on the T4

STATISTICS

Wilcoxon signed-rank + Holm · exact McNemar · TOST for equivalence · Clopper–Pearson intervals · bootstrap CIs

SOURCES Radford et al. (2023) · S. Chen et al. (2022) · Saeki et al. (2022), *UTMOS* · Wilcoxon (1945) · Schuirmann (1987)

CONTRIBUTION & SUMMARY

WHAT THIS STUDY ADDS

Within the sources reviewed, no method combines all three for flow-matching TTS.

	PRIOR WORK	THIS STUDY
01 **When to hand over**	One switching time for every utterance: fixed at 0.25 in DMOSpeech 2, preset in T-Stitch.	Chosen **per utterance**, from signals the sampler already computes: the prompt's SNR and the endpoint-stabilization score.
02 **How sure we can be**	Adaptive rules use learned or hand-tuned thresholds with **no bound** on quality loss. Calibrated bounds exist, but for exit layers and ASR hypotheses.	A **Learn-Then-Test** threshold: added failures ≤ $\alpha = 0.05$ with probability ≥ 90%, checked on held-out speakers.
03 **What it costs**	Some need extra training, or a quality estimator run on a full output.	**No training and no extra forward pass.** With no certified threshold it falls back to the released sampler, so it never costs more.

NOT CLAIMED A proven advantage of the hybrid over its parts, any change to duration prediction, or any new model training.

FROM Proposal, Section 2.8, Section 3.7 and Propositions 3.2 and 3.4

APPENDIX D.1–D.3 & EXP-017 · PRELIMINARY & REHEARSAL RUNS (TESLA T4)

FIXED-SCHEDULE TRADE-OFF & BASELINE CHECKS

D.1 · FIXED SWITCHING TIMES · 500 SPEAKERS × CLEAN & 15 DB NOISY · TESLA T4

SWITCH	PASSES	CLEAN (500)	NOISY (500)	ALL (1,000)
s = 0.07 (Step 4)	13	12 (2.4%)	**322 (64.4%)**	334 (33.4%)
s = 0.10 (Step 5)	15	9 (1.8%)	**269 (53.8%)**	278 (27.8%)
s = 0.15 (Step 6)	17	5 (1.0%)	**247 (49.4%)**	252 (25.2%)
s = 0.25 · ref (Step 8)	21	0	0	0

Hindsight per-input selection needs **14.85 of 21 passes (−29.3%)** on 1,000 instances (confirmed in **EXP-017**: **29.7%** 16-grid & **33.7%** 32-grid). Across all **Fixed Steps 1–8 (EXP-018, n = 360)**, clean speech already converges at **Step 2 (9 NFE, 1.44 s, 2.64% WER)** while noisy speech collapses across Steps 1–4 (**280.2%–24.7% WER**) until **Step 7 (19 NFE**, TOST $p = 0.0047$), expanding oracle headroom to **39.3%–43.5%**.

D.2 · STUDENT RESTART LEVEL & PAPER 14 + 2 BASELINES

In EXP-017 on the Tesla T4 (n = 360), the paper's **14 + 2 (j = 2, 32 passes)** added **140/360 failures** (noisy WER 0.342 vs 0.076) and **Algorithm 2** (no re-noising) added **172/360**; 32-grid s = 0.25 (33 passes) matched 16-grid Step 8 at **+57% cost**. Keeping **j = 1 (τ = 0.0761, 3 passes)** is empirically justified.

D.3 · AUDITED STABILIZATION RULE & BINARY NOISE FLOOR

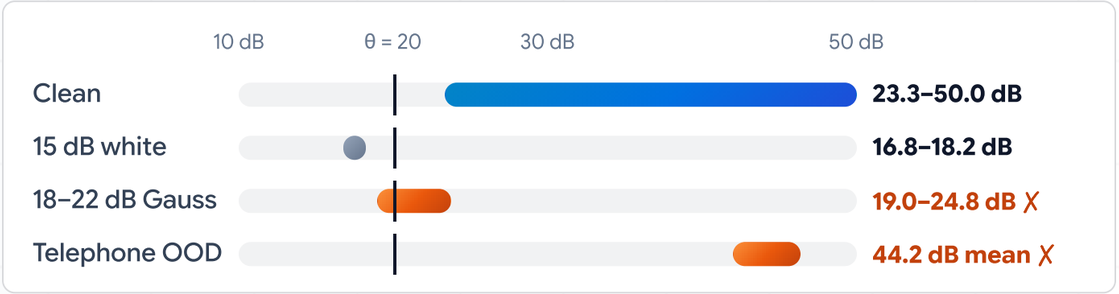
Post-proposal T4 audits (**AUDIT-001, EXP-012, EXP-017**) showed Definition 3.4's low-drift rule ($d_j \le \lambda$) certifies **0.0% saving on the 16-grid** (1.8% on 32-grid) under LTT. Moreover, comparing two equal-quality references still flagged **81/360 (22.5%) binary "failures"** from student seed noise, motivating paired TOST evaluation.

FROM Proposal Appendix D.1–D.3 · AUDIT-001 & EXP-017 (Tesla T4, n = 1,000 & n = 360) · development evidence, not unseen held-out validation

APPENDIX D.4–D.5 & EXP-011–015 · SIGNAL DIAGNOSTIC (TESLA T4)

WHY SNR GATES & LOW-DRIFT STOPPING BREAK

PROMPT SNR · CLEAN VS. MODERATE & OOD CORRUPTIONS (EXP-011 & EXP-012)



While θ = 20 dB separated clean from 15 dB white noise, it **missed 18–22 dB noise (86/360 = 23.9% failures, 4.8× over 5% budget)** and was fooled by bandpass **telephone audio (44.2 dB)** and babble (23.6 dB), while adding **1.85 ms** waveform overhead.

EXP-012 DISCOVERY · NOISE CAUSES PREMATURE STABILIZATION

On clean prompts, single-step d_4 had AUC ≈ **0.47** (2–3 failures). Across noise levels (n = 360), degraded prompts **stabilize prematurely**: median d_4 drops from **0.61 (clean)** to **0.17 (15 dB noisy)** as teacher guidance contrast collapses (ρ = 0.89). Thus $d_i \le \lambda$ stops **noisy prompts first**.

EXP-013–018 RESOLUTION · CUMULATIVE TRAJECTORY ACTION

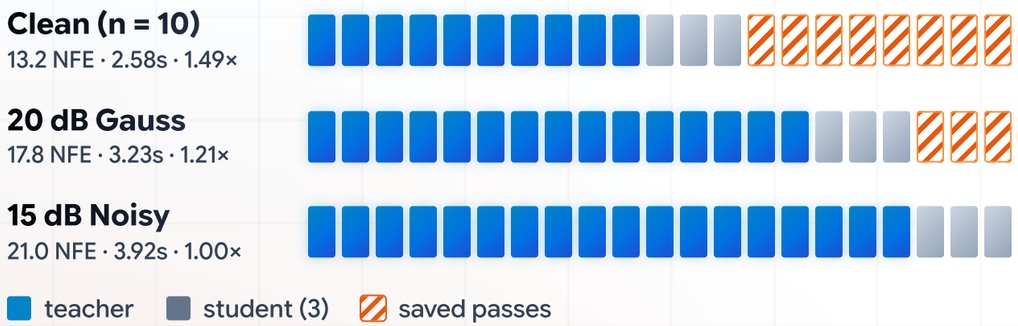
Eight state-altering operators (**DEH, TCR, cDEH, SEA, SNNG, SCG, PIEC**) failed routing criterion C3, and 1-cut SNR gates collapsed to **1.5% saving** when Step 1 was included. Summing path length $A_i = \sum_{j=1}^{i} d_j \ge \hat{\lambda}_i$ flips the direction, smooths step jitter, and **removes Prompt SNR**.

FROM Proposal Appendix D.4–D.5 · EXP-011, EXP-012 & EXP-013–015 (Tesla T4, n = 360 & OOD n = 100) · development evidence

APPENDICES B–C & EXP-018 · LIVE GPU EXECUTION & RISK CONTROL

LIVE T4 HANDOVER SPEEDUP & RISK CALIBRATION

LIVE TESLA T4 EARLY EXIT · STAC (n = 30, max|diff| = 0.0)



Online GPU stopping saves **37.1% NFE (1.49× E2E, 1.68× teacher)** on clean speech, **15.2% (1.21×)** at 20 dB, and falls back **100% to Step 8** at 15 dB (**0.12 ms** gate overhead).

CALIBRATION · APPENDIX B SIMULATION & 1D-CRC VERIFICATION

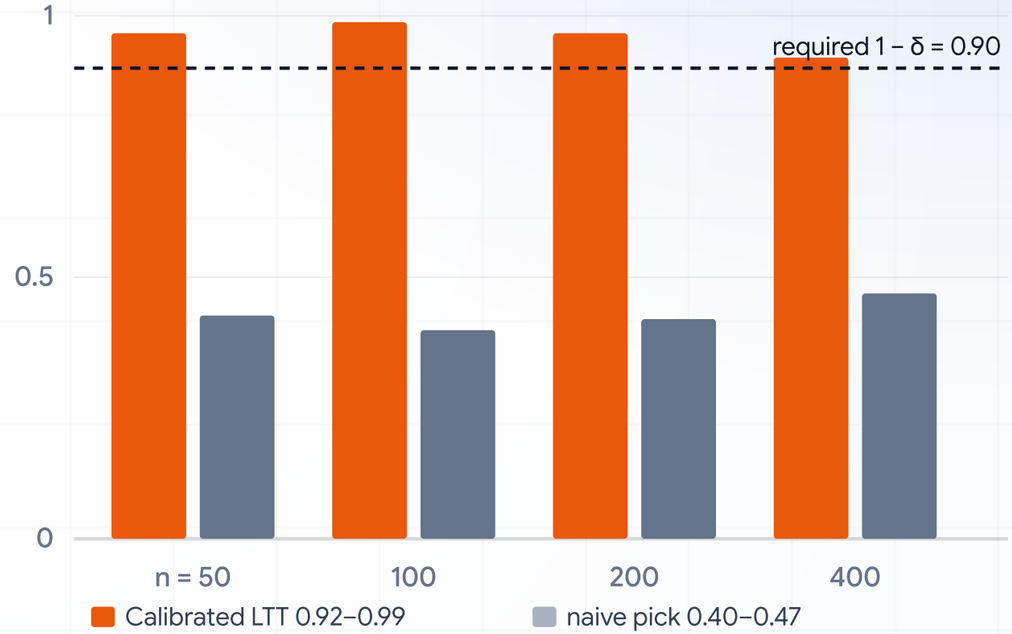


Chart: Appendix B simulated stress test. On T4 development runs (EXP-018), **1D-CRC-STAC (α = 0.05)** held WER to 0.0670 (vs 0.0679 ref) at 17.67 NFE, and LTT certified **9/360 (2.5% ≤ 5%)** without SNR.

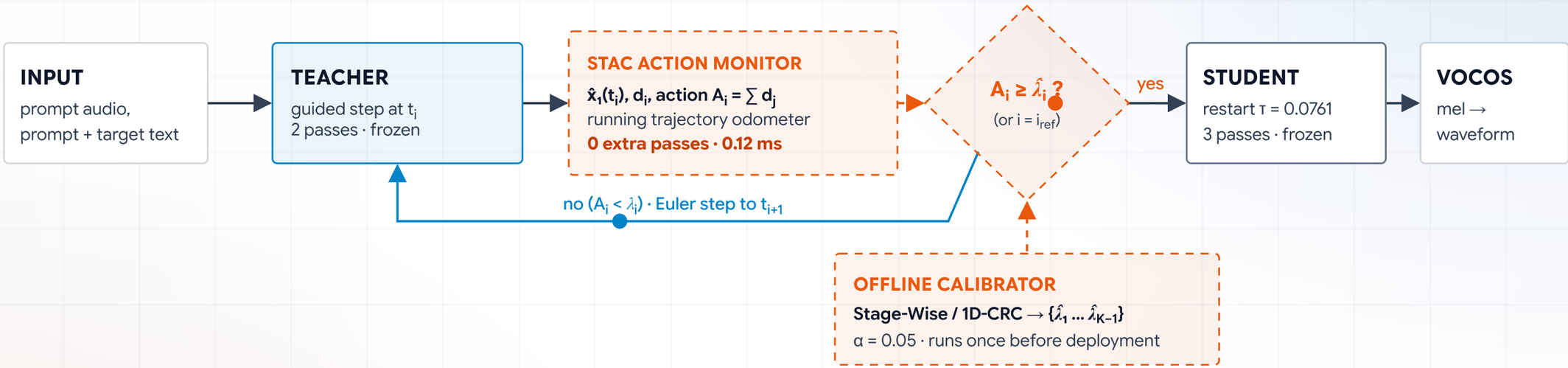
FROM Proposal Appendices B–C & EXP-018 Live Tesla T4 Telemetry (live_swsr_t4.csv, n = 30) · development evidence

APPENDIX E.1 · NOVEL HANDOVER METHOD (EXP-018)

STAC: STAGE-WISE TRAJECTORY ACTION CASCADE

~~PROMPT SNR (Ŝ, gate 0) — ELIMINATED~~
0 external heuristics · saves 1.85 ms · immune to OOD bandpass

released, frozen
added by STAC (this study)



CLEAN PROMPT · UTTERANCE #1
Strong guidance builds action fast: ***A_4* = 3.65 ≥ $\hat{\lambda}_4$ (1.59)** → exits at **Step 4 (13 NFE, −38%)**.

20 DB GAUSS · UTTERANCE #1
A_4 = 1.52 < 1.59 (waits!), accumulates to ***A_7* = 2.06 ≥ $\hat{\lambda}_7$ (1.93)** → exits at **Step 7 (19 NFE, −10%)**.

15 DB NOISY · UTTERANCE #1
Prematurely flat (A_7 = 1.76 < 1.93) → stays in loop to **Step 8 ref (21 NFE, 0 added loss)**.

FROM EXP-018 (run_swsr_handover_prelim.py, steps1to8.csv) · Pure switching-time policy: frozen teacher, student, duration & τ = 0.0761 · development evidence

APPENDIX E.2 · EMPIRICAL BENCHMARK COMPARISON (TESLA T4)

STAC RESULTS ACROSS GRIDS, COHORT & OOD NOISE

HANDOVER POLICY (N = 360, 5-FOLD CV)	SNR?	16-GRID (1..8) NFE	E2E TIME · SPEEDUP	WER (TOST P)	SIM	32-GRID NFE (SAVED)
STAC ($A_i \geq \hat{\lambda}_i$) · Winner	**No · 0.12 ms**	**16.82 (−19.9%)**	**2.60s · 1.25× (1.79× RT)**	**0.0687 (1.2e−7 ✓)**	**0.8094**	**26.18 (−20.7%)**
1D-CRC-STAC (α = 0.05 certified)	**No · 0.12 ms**	**17.67 (−15.8%)**	**2.73s · 1.19× (1.71× RT)**	**0.0670 (2.6e−8 ✓)**	**0.8101**	**27.99 (−15.2%)**
Stage-Wise Only · no sum ($d_i \geq \hat{\lambda}_i$)	**No · 0.12 ms**	16.91 (−19.5%)	2.61s · 1.24× (1.78× RT)	0.0684 (4.4e−6 ✓)	0.8097	28.35 (−14.1%)
Action Sum Only · 1 gate ($A_i \geq \hat{\lambda}$)	**No · 0.12 ms**	20.30 (−3.3%)	3.13s · 1.03× (1.49× RT)	0.0670 (fails all1000)	0.8103	30.27 (−8.3%)
Stage-Wise SNR Only ($\hat{S} \geq \hat{\theta}_i$)	Yes · 1.85 ms	18.13 (−13.7%)	2.80s · 1.15× (1.66× RT)	0.0682 (1.5e−20 ✓)	0.8098	27.94 (−15.3%)
1-Gate SNR / SNR×Action / Def. 3.4	Yes · 1.97 ms	20.15–21.00 (0–4%)	3.11–3.23s · 1.00–1.04×	0.0679–0.0685	0.8103	29.23–32.79 (0–11%)
Fixed Step 8 Ref · (Teacher-Only 32)	—	21.00 · (64 NFE)	3.23s · (9.65s, 3.7× slower)	0.0679 · (0.815 T-32)	0.8104	33.00 (5.03s)

500-SPEAKER COHORT · ALL1000

200 cal → 200 test speakers (n = 400): **STAC & 1D-CRC-STAC reach 16.63 NFE (−20.8%, 2.57s, 1.25×)**, Capped WER 0.1223 (−0.0008 vs ref), TOST p = 0.0018.

ZERO-SHOT OOD NOISE · EXP-011

Across babble, reverb & telephone (n = 100): **STAC saves 32.7% (14.14 NFE) vs 17.1% (17.40 NFE)** for SNR Cascade (ΔWER −0.0021), catching 44.2 dB telephone audio.

DIAGNOSTIC & NEXT STEP

On 14-stage g32_all15, ultra-early t_2 = 0.0048 caused 1–2 Whisper repetition outliers (Capped WER 0.078, Raw 0.152 → set $t_k \geq 0.019$). **Frozen unseen-speaker test (EVAL-001/002) pending.**

FROM EXP-011 to EXP-018 (Tesla T4, steps1to8.csv & swsr_results.json) · development evidence, pending unseen held-out evaluation & adviser review